

Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pe nding-x7q2-rerun

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pending

Profile snapshot

- **Primary site:** bone marrow (hematologic)
- **Histology:** Acute myeloid leukemia, myelodysplasia-related (AML-MR), relapsed/refractory; transformed from prior MDS (RCMD, IPSS-R high/very-high, diagnosed ~2019); with extramedullary myeloid sarcoma
- **Stage:** Relapsed/refractory AML-MR. Marrow ~85% blasts (June 2026); peripheral-blood blasts 58->82% (late June 2026); extramedullary right-thigh myeloid sarcoma on MRI (May 2026, adverse feature). Refractory to FLAG-IDA + venetoclax reinduction. Late relapse ~6 years after matched-sibling PBSC allo-HCT (~2020); pattern consistent with graft-versus-leukemia escape rather than graft failure.
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** female
- **Biomarkers:** TP53 aberrant, pending confirmation (NOT resolved in either direction). Pathogenic TP53 mutation reported on 2019 diagnostic NGS; 2026 relapse NGS reported NEGATIVE, but on a limited assay (LOD ~2% VAF, no copy-number/CNV detection, not MRD-grade). Discrepancy most plausibly subclonal loss, copy-loss, or a re-review artifact rather than true reversion. Working call: TP53-aberrant pending confirmation. (ngs_pending); TP53 Y220C (specific variant) unknown / to be determined — Y220C status not established; therapy handle (rezatapopt) is contingent on it (ngs_pending); HLA-A02:01 *unknown / pending retrieval*. *High-resolution HLA typing was performed ~2020 but the allele-level result is not in the records on hand (placeholder in the EHR).* (unknown); HLA-A24:02 unknown / pending (unknown); HA-1 / HA-2 minor histocompatibility antigens (recipient and sibling donor) unknown / pending for BOTH patient and sibling donor. HLA-identical siblings can still differ at HA-1/HA-2. (unknown); CD33 positive (blasts CD33+ by flow cytometry); CD123 positive (blasts CD123+ by flow cytometry); AML immunophenotype (supporting) CD34+ (increased), CD13+, CD15+ (subset), CD117+ (variably increased), HLA-DR (variably decreased), CD38 decreased; negative CD4/CD5/CD7/CD14/CD16/CD19/CD56/CD64; FLT3 negative / no mutation (2019 diagnostic NGS); IDH1 negative / no mutation (2019 diagnostic NGS); IDH2 negative / no mutation (2019 diagnostic NGS); JAK3 variant of uncertain significance (VUS) — 2019 (unknown); SMC3 variant of uncertain significance (VUS) — 2019 (cohesin-complex gene) (unknown); Cytogenetics / karyotype Complex, adverse-risk. 2019: complex karyotype with del(5q), del(7q), del(20q12), der(17)t(13;17) [17p], monosomy 13, others. 2026 relapse: gain of 5' MECOM, del(5q), del(7q), KMT2A amplification (4 copies), del(8cen), del(16q22). Shared del(5q)/del(7q) + TP53 founder across timepoints indicates evolution of the ORIGINAL clone transformed to AML (not a new/secondary or donor-derived leukemia). (ngs_pending)
- **Efficacy/toxicity weight:** 0.75

6 ranked therapeutic options across 4 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Allograft Hct2 interventions

1. Iomab-B (131I-apamistamab) anti-CD45 radioimmunotherapy conditioning + second allo-HCT (HCT2 platform)

Likelihood of effect: Moderate: the only randomized signal in the dossier (SIERRA dCR 17.1% vs 0%, $p < 0.0001$), but read indirectly onto a prior-HCT relapse it did not enroll.

Toxicity burden: High (grade 3+ TRAEs 59.7%, at parity with conventional care — conditioning cytopenias, infection, mucositis; second-transplant TRM uncharacterized)

Counter-productive MoA:Low (Radioconjugate marrow injury could compromise engraftment if the HCT2 platform is not secured; no persona raised a mechanism objection.)

Overall:The curative backbone all five personas endorsed — the randomized way to carry active disease into HCT2, gated on a fitness workup that has not yet been done.

Key references: PMID 39298738 · NCT02665065 · NCT07157514

Evidence in detail

lomab-B (131I-apamistamab) anti-CD45 radioimmunotherapy conditioning + second allo-HCT (HCT2 platform)

Gyurkocza et al. *J Clin Oncol* 2025

- **Disease context:**Active relapsed/refractory AML, age ≥ 55 , planned allo-HCT (2L+) — Active R/R AML age ≥ 55 fit for allo-HCT; ITT included crossover (52% of conventional-care arm received 131I-apamistamab). Enrolled patients heading to a first allo-HCT.
- **n:** 153
- **Toxicities:**Any treatment-related AE (grade ≥ 3): 59.7% (n=76)
- **Efficacy:**CR_rate 17.1% (durable CR ≥ 180 d) (95% CI 9.4-27.5); HR_OS 0.99 (95% CI 0.7-1.41); HR_EFS 0.23 (95% CI 0.15-0.34)

Hla A0201 interventions

1. HA-1-specific CD8+/CD4+ TCR-T (Bleakley platform; TSC-100 / BSB-1001) *(conditional on hla_a0201 positive)*

Likelihood of effect: Assuming A*02:01 + HA-1 mismatch confirm: uncertain but curative-intent — 4/9 CR in Krakow 2024 (n=9 feasibility, no comparator). Negative typing forecloses it entirely.

Toxicity burden: Moderate (lymphodepletion neutropenia, grade 3-5 infection; mild grade 1-2 GVHD; no CRS, no ICANS across 9 patients)

Counter-productive MoA:Low (TCR-T kinetics may lag an 85%-blast marrow and antigen-loss escape is possible, but lineage-restricted killing limits off-target collateral.)

Overall:The only lever that answers the referral's own question — a clean-safety, curative-intent graft-versus-leukemia consolidation, live only if the pending typing confirms and delivered after HCT2.

Key references: PMID 38683966 · NCT05473910 · NCT03326921

Evidence in detail

HA-1-specific CD8+/CD4+ TCR-T (Bleakley platform; TSC-100 / BSB-1001)

****Krakow et al. *Blood* 2024****

- **Disease context:**Recurrent leukemia/myeloid neoplasm after allo-HCT (2L+) — Recurrent AML/ALL/MDS/JMML after allo-HCT; HLA-A*02:01-positive with recipient HA-1(H)-positive / donor HA-1-negative mismatch. n=9 treated.
- **n:** 9
- **Toxicities:**CRS (grade any): 0% (n=9); ICANS (grade any): 0% (n=9); GVHD (acute) (grade 1-2): (n=9); Neutropenia (grade >=3): (n=9); Infection (grade >=3): (n=9)
- **Efficacy:**CR_rate 4/9 patients; AE_rate 0 DLTs

Cd123 interventions

1. Tagraxofusp (SL-401, CD123-directed IL-3/diphtheria-toxin fusion)

Likelihood of effect: Low-to-uncertain for AML: headline 72% CR is first-line BPDCN cross-tumor (Pemmaraju 2019); no AML efficacy data and CD123 density still pending.

Toxicity burden: High (capillary leak syndrome with treatment-related deaths, transaminitis ALT 64% / AST 60%, hypoalbuminemia 55%)

Counter-productive MoA:Low (No clear mechanistic antagonism; the efficacy risk is cross-tumor non-transfer from BPDCN, not a counter-productive vector against the goal.)

Overall:The only labeled CD123 agent with a real safety algorithm, but its headline efficacy is BPDCN cross-tumor and its capillary-leak deaths demand organ-function labs first.

Key references: PMID 31018069 · NCT06561152

2. Pivekimab sunirine (IMGN632, CD123 ADC)

Likelihood of effect: Moderate as debulking: own R/R AML data (ORR 21%, 95% CI 8-40; cCR 17%), but single-agent depth is modest against an 85%-blast marrow.

Toxicity burden: High (veno-occlusive disease at higher doses, one treatment-related death; RP2D: febrile neutropenia 10%, infusion reactions 7%)

Counter-productive MoA:High (Veno-occlusive disease from the ADC ahead of HCT2 could sink the very transplant the debulking is meant to enable.)

Overall:The CD123 ADC with the best own-disease data, held under a live, conditional VOD veto — it reinstates as a debulking bridge the moment hepatic and albumin baselines clear it.

Key references: PMID 38423051 · NCT03386513 · PMID 29661755

3. Flotetuzumab (MGD006, CD123xCD3 DART)

Likelihood of effect: Uncertain and not enrollable: 30% ORR in refractory AML (Uy 2021), TP53 signal from a post-hoc correlative subgroup (Vadakekolathu 2020); program discontinued.

Toxicity burden: Moderate (cytokine release syndrome / infusion reactions, largely grade 1-2 with step-up dosing, dexamethasone, tocilizumab)

Counter-productive MoA: **Moderate** (CD123xCD3 redirection needs fit autologous T cells; exhaustion after transplant and FLAG-IDA may blunt it, and the TP53 rationale is post-hoc.)

Overall: A discontinued drug carrying a live mechanism signal — its TP53-inflamed-marrow rationale is post-hoc, and the reason to chase an open CD123xCD3 construct rather than this one.

Key references: PMID 32929488 · PMID 33057635 · NCT02152956

Evidence in detail

Tagraxofusp (SL-401, CD123-directed IL-3/diphtheria-toxin fusion)

Pemmaraju et al. *N Engl J Med* 2019

- **Disease context:** **Blastic plasmacytoid dendritic-cell neoplasm (BPDCN), CD123+** (any) — CD123+ BPDCN; 29 previously untreated at 12 µg/kg (primary-outcome cohort) plus 15 previously treated. Not the patient's disease.
- **n:** 47
- **Toxicities:** **Capillary leak syndrome** (grade any): 19% (n=47); **ALT increased** (grade any): 64% (n=47); **AST increased** (grade any): 60% (n=47); **Hypoalbuminemia** (grade any): 55% (n=47); **Peripheral edema** (grade any): 51% (n=47); **Thrombocytopenia** (grade any): 49% (n=47)
- **Efficacy:** **CR_rate** 72%; **ORR** 90%; **OS_rate** 59% at 18 mo; **OS_rate** 52% at 24 mo; **ORR** 67%; **OS** 8.5 months

Pivekimab sunirine (IMGN632, CD123 ADC)

Daver et al. *Lancet Oncol* 2024

- **Disease context:** **Relapsed/refractory AML, CD123+** (2L+) — CD123+ R/R AML, ECOG 0-1; first-in-human dose-escalation/expansion, RP2D expansion cohort n=29 (whole study n=91).
- **n:** 91
- **Toxicities:** **Febrile neutropenia** (grade ≥3): 10% (n=29); **Infusion-related reaction** (grade ≥3): 7% (n=29); **Anaemia** (grade ≥3): 7% (n=29); **Veno-occlusive disease** (grade ≥3): (n=68); **Treatment-related death** (grade 5): (n=68)
- **Efficacy:** **ORR** 21% (95% CI 8-40); **CR_rate** 17% (composite CR) (95% CI 6-36)

Flotetuzumab (MGD006, CD123xCD3 DART)

Uy et al. *Blood* 2021

- **Disease context:** **Primary induction failure / early-relapse R/R AML (2L+)** — Primary induction failure or early relapse (<6 mo) R/R AML; benefit enriched in an immune-infiltrated tumor microenvironment. RP2D PIF/ER subset n=30 (whole study n=88).

- **n:** 88
- **Toxicities:**CRS / infusion-related reaction (grade any): (n=88); **CRS / infusion-related reaction** (grade ≥ 3): (n=88)
- **Efficacy:**CR_rate 26.7% (CR/CRh); **ORR** 30%; **OS** 10.2 months; **OS_rate** 75% at 6 mo; **OS_rate** 50% at 12 mo

Cd33 interventions

1. Gemtuzumab ozogamicin (CD33 ADC)

Likelihood of effect: Low for this stratum: tier-1a Hills meta shows OR 0.99 (95% CI 0.83-1.18) — no survival benefit in adverse cytogenetics, the patient's exact subgroup.

Toxicity burden: Moderate (grade 3+ hyperbilirubinemia 23%, transaminitis 17%, veno-occlusive-disease class risk ahead of HCT2)

Counter-productive MoA:Moderate (P-glycoprotein efflux of the calicheamicin payload in adverse-karyotype blasts blunts CD33-ADC delivery — the subgroup where its own data show no benefit.)

Overall:The on-label, guideline-listed CD33 handle whose own tier-1a evidence shows no survival benefit in this complex/TP53-aberrant karyotype — familiar, but pointed the wrong way.

Key references: PMID 11432892 · PMID 25008258 · PMID 22482940

Evidence in detail

Gemtuzumab ozogamicin (CD33 ADC)

Sievers et al. *J Clin Oncol* 2001

- **Disease context:**CD33+ AML in untreated first relapse (2L+) — CD33+ AML in untreated first relapse, no antecedent hematologic disorder; median age 61. Pooled across three open-label trials.
- **n:** 142
- **Toxicities:**Hyperbilirubinemia (grade ≥ 3): 23% (n=142); **Transaminase elevation** (grade ≥ 3): 17% (n=142); **Infection** (grade any): 28% (n=142); **Mucositis** (grade ≥ 3): 4% (n=142); **Nausea/vomiting (severe)** (grade ≥ 3): 11% (n=142)
- **Efficacy:**CR_rate 30% (CR+CRp)

Hills et al. *Lancet Oncol* 2014

- **Disease context:**Adult AML (non-APL), front-line induction (1L) — 3325 adults with AML (excluding APL) in five randomized induction trials; cytogenetic risk subgroups analyzed, including an adverse-cytogenetics stratum.
- **n:** 3325
- **Toxicities:**Early death (grade 5): (n=3325)
- **Efficacy:**HR_OS 0.9 OR (5y OS) (95% CI 0.82-0.98); **other** 0.81 OR (relapse) (95% CI 0.73-0.9); **HR_OS** 0.99 OR (95% CI 0.83-1.18); **HR_OS** 0.47 OR (95% CI 0.31-0.73)

Castaigne et al. *Lancet* 2012

- **Disease context:** **De novo AML, front-line (1L)** — Previously untreated de novo AML age 50-70 (excludes APL); CD33 expression not required for entry. 280 randomized.
- **n:** 280
- **Toxicities:** **Persistent thrombocytopenia** (grade any): 16% (n=139); **Death from toxicity** (grade 5): (n=139)
- **Efficacy:** **HR_EFS** 0.58 (95% CI 0.43-0.78); **HR_OS** 0.69 (95% CI 0.49-0.98); **HR_RFS** 0.52 (95% CI 0.36-0.75); **CR_rate** 81% vs 75%